

Supplemental Material: Momentum correlations of the Hawking effect in a quantum fluid

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(Dated: September 21, 2026)

MEAN-FIELD DYNAMICS AND POLARITON KINEMATICS

Polaritons are quasiparticles of mass m^* resulting from the strong coupling of cavity photons and quantum well excitons. Under coherent excitation by a pump $F(x, t)$ near resonance with the cavity, the mean field of polaritons can be described by the driven-dissipative Gross-Pitaevskii equation [1]

$$i\partial_t\psi(x, t) = F(x, t) + \left(\omega_0 - \frac{i\gamma}{2} - \frac{\hbar\partial_x^2}{2m^*} + g|\psi(x, t)|^2 + V(x) \right) \psi(x, t). \quad (S1)$$

Here ω_0 is the polariton frequency at wavenumber $k = 0$, γ the (mostly photonic) decay rate, $g > 0$ the repulsive interaction strength and $V(x)$ the potential reigning in the cavity.

We consider a monochromatic pump $F(x, t) = F(x)e^{-i\omega_p t}$. It is useful to decompose the polariton field as $\psi(x, t) = \sqrt{n(x, t)}e^{i(\theta(x, t) - \omega_p t)}$. The density n and the phase θ satisfy the continuity equation

$$\partial_t n + \partial_x(nv) = -\gamma n + 2\sqrt{n} \operatorname{Im} F e^{-i\theta}, \quad (1)$$

and the Euler-like equation (via its gradient)

$$\partial_t \theta + gn - \delta_0 + \frac{m^* v^2}{2\hbar} + V - \frac{\hbar}{2m^*} \frac{\partial_x^2 \sqrt{n}}{\sqrt{n}} + \frac{\operatorname{Re} F e^{-i\theta}}{\sqrt{n}} = 0, \quad (2)$$

where $v = \hbar\partial_x\theta/m^*$ is the fluid velocity and $\delta_0 = \omega_p - \omega_0$ the bare detuning.

For a plane-wave pump $F(x) = \sqrt{I}e^{ik_p x}$, the stationary homogeneous solution $\psi = \sqrt{n}e^{i(k_p x - \omega_p t + \theta_0)}$ satisfies

$$\left[(gn - \delta)^2 + \frac{\gamma^2}{4} \right] n = I, \quad (S2)$$

where $\delta = \delta_0 - \hbar k_p^2/(2m^*) + V$, and $\theta_0 = \arctan[\gamma/(2(\delta - gn))]$. This nonlinear relation exhibits optical bistability under appropriate detuning conditions [2], as shown in Fig. 1.

BOGOLIUBOV-DE GENNES ANALYSIS AND DISPERSION RELATION

Collective excitations are studied by linearizing around the steady state: $\psi(x, t) = [\sqrt{n} + \varphi(x, t)e^{-\gamma t/2}]e^{i(k_p x - \omega_p t)}$, with

$$\varphi(x, t) = \sum_q \left[u_q e^{i(qx - \omega t)} \hat{a}_q + v_q^* e^{-i(qx - \omega t)} \hat{a}_q^\dagger \right], \quad (S3)$$

where q is the relative wave number with respect to the pump ($k = k_p + q$), and u_q, v_q are complex mode functions obeying the Bogoliubov-de Gennes equations in the driven-dissipative regime [3].

In the laboratory frame, the dispersion reads

$$\omega^\pm = vq \pm \sqrt{\left(\frac{\hbar q^2}{2m^*} - \delta(k_p) + 2gn \right)^2 - (gn)^2}. \quad (S4)$$

Modes on ω^+ have positive symplectic norm $Q = \int dx (|u_q|^2 - |v_q|^2)$; those on ω^- have negative norm [4].

Sound speed and healing length. Because $gn(x) \gtrsim \delta(k_p)(x)$ in both regions, the dispersion is parabolic even at low wavenumbers. The sound cones are therefore not given by the standard $c_s = \sqrt{\hbar gn/m^*}$ but by $c_B = \sqrt{\hbar(2gn - \delta(k_p))/m^*}$ [5]. The healing length $\xi = \sqrt{\hbar/m^*(2gn - \delta(k_p))}$ sets the microscopic scale controlling density modulation relaxation and the low- $|q|$ Bogoliubov dispersion [4].

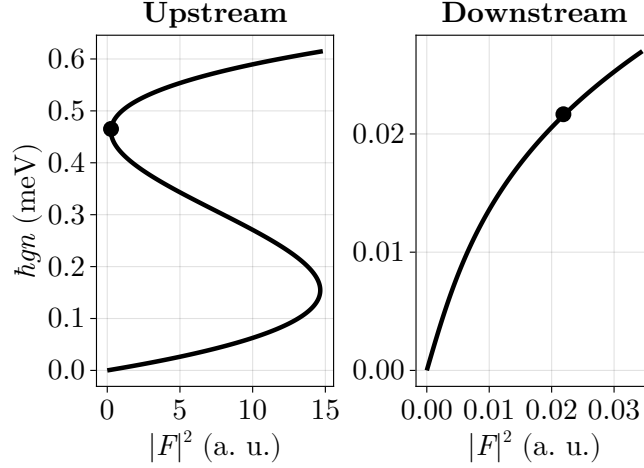


FIG. 1. Bistability curves Eq. (S2). The dot denotes the working point

Effective Klein–Gordon field. The perturbation φ obeys a Klein–Gordon equation on the curved spacetime defined by the Painlevé–Gullstrand line element

$$ds^2 = (c_B^2(x) - v^2(x))dt^2 - 2v(x)dxdt + dx^2, \quad (3)$$

with an effective mass m_{det} set by $|F_p|^2$ and $\delta(k_p)$:

$$\frac{1}{\sqrt{|g|}}\partial_\mu(\sqrt{|g|}g^{\mu\nu}\partial_\nu\varphi) - \frac{m_{\text{det}}^2}{\hbar^2}\varphi = 0. \quad (4)$$

KILLING HORIZON AND MODE STRUCTURE

The inhomogeneous pump creates a profile where $v(x) < c_B(x)$ upstream ($x - x_d < 0$) and $v(x) > c_B(x)$ downstream ($x - x_d > 0$). In the subcritical region, the positive- and negative-norm branches occupy positive and negative frequencies respectively. In the supercritical region, ω^- is drawn to positive frequencies (up to ω_{max}) and ω^+ to negative frequencies, enabling the excitation of negative-energy modes ($\text{sign}(E) = \text{sign}(\omega) \times \text{sign}(Q)$ [6]). This unambiguously signals the formation of a Killing horizon [5].

The upstream working point is chosen so that gn is as close as possible to $\delta(k_p)$ [black dot in 1], reaching the optimal configuration for spontaneous Hawking emission [7]. In the downstream region, $\delta(k_p)$ is small (no hysteresis); $|F|^2$ is chosen high to maintain finite density support while minimizing phase-mismatch undulations.

Given the dispersion on either side of the horizon, we distinguish three propagation regimes. When $\omega < \omega_{\text{min}}$, propagation is not allowed upstream and there are four propagating modes downstream. When $\omega_{\text{min}} < \omega < \omega_{\text{max}}$, there are two propagating modes upstream and four downstream—this is the Hawking window. When $\omega > \omega_{\text{max}}$, there are two propagating modes upstream and two downstream.

The scattering matrix (Eq. (3) of the main text) connects three incoming modes to three outgoing modes in the Hawking window. Within Bogoliubov theory, for modes of negative norm such as dn, the relevant operator appearing in observables is $\hat{b}_{\text{dn}}$ (not its Hermitian conjugate) due to the anomalous sign in the commutation relations. The dominant pair correlation is therefore anomalous: $\langle \hat{b}_{\text{HR}} \hat{b}_{\text{dn}} \rangle \neq 0$. In contrast, normal correlations such as $\langle \hat{b}_{\text{HR}} \hat{b}_{\text{dn}}^\dagger \rangle$ are generically suppressed unless induced by temperature or nonadiabatic effects.

Analytical predictions for $g_2(k, k')$ show that the HR–dn* channel dominates and survives far beyond regimes where thermal occupation numbers obscure diagonal terms [8–11]. These correlations are enhanced at low frequencies, peak near ω_{min} , and decay toward ω_{max} , reflecting the frequency dependence of the Bogoliubov coefficients [12, 13].

To identify correlation traces in a (k, k') plot, one tracks the dispersion (S4) at fixed ω : the allowed momenta in each asymptotic region define loci $k_i(\omega)$ that are real within $\omega_{\text{min}} < \omega < \omega_{\text{max}}$. The sign of the Bogoliubov norm distinguishes positive- and negative-energy branches.

THE TRUNCATED WIGNER APPROXIMATION

The Truncated Wigner Method (TWM) allows one to efficiently simulate quantum expectation values and has been extensively employed in studies of polaritons [14]. The master equation for this system in the Wigner representation may be approximated by a Fokker-Planck equation if the condition $\gamma \ll g/\Delta x$ is satisfied, where Δx is the grid spacing used in the simulation. The corresponding Langevin equations in the Ito sense are

$$i d\psi(x, t) = A\{\psi\}(x) dt + \sqrt{\frac{\gamma}{2\Delta x}} dW(x, t), \quad (S5)$$

where $A\{\psi\}(x)$ denotes the right-hand side of Eq. (S1), and $dW(x, t)$ is a complex Wiener process whose increment $\xi(x) = \int_{t_0}^{t_0+\Delta t} dW(x, t)$ satisfies $\langle \xi(x) \rangle = 0$ and $\langle \xi(x) \xi^*(x') \rangle = \delta_{x,x'} \Delta t$.

The initial conditions are sampled from the Wigner distribution of the vacuum (initially empty cavity): $\psi(x, 0)$ is a complex Gaussian variable with $\langle \psi(x, 0) \rangle = 0$ and $\langle \psi(x, 0) \psi^*(x', 0) \rangle = \delta_{x,x'}/(2\Delta x)$. Averages over the stochastic process correspond to symmetrically ordered expectation values:

$$\langle O_1 \cdots O_N \rangle_W = \frac{1}{N!} \sum_P \langle \hat{O}_{P(1)} \cdots \hat{O}_{P(N)} \rangle, \quad (5)$$

where the sum runs over all permutations P .

Normally ordered correlation functions. The first-order correlation is

$$G_1(x, x') = \langle \hat{\psi}^\dagger(x) \hat{\psi}(x') \rangle = \langle \psi^*(x) \psi(x') \rangle_W + \frac{\delta_{x,x'}}{2\Delta x}, \quad (6)$$

with populations $n(x) = G_1(x, x)$. The second-order correlation is

$$\begin{aligned} G_2(x, x') &= \langle \hat{\psi}^\dagger(x') \hat{\psi}^\dagger(x) \hat{\psi}(x) \hat{\psi}(x') \rangle \\ &= \langle |\psi(x)|^2 |\psi(x')|^2 \rangle_W \\ &\quad - \frac{1 + \delta_{x,x'}}{2\Delta x} \left\langle |\psi(x)|^2 + |\psi(x')|^2 - \frac{1}{2\Delta x} \right\rangle_W. \end{aligned} \quad (7)$$

The normalized second-order correlation $g_2(x, x') = G_2(x, x')/[n(x)n(x')]$ yields the real-space signatures shown in Fig. 1(e) of the main text.

MOMENTUM-SPACE CORRELATIONS: WINDOWED FOURIER TRANSFORM

The windowed Fourier transform of the field is

$$\hat{\psi}_a(k, t) = \int_0^L dx e^{-ikx} w_a(x) \hat{\psi}(x, t), \quad (8)$$

where $w_a(x)$ is a window function localized in region $a \in \{u, d\}$. The commutation relations become

$$[\hat{\psi}_a(k), \hat{\psi}_b^\dagger(k')] = f_{ab}(k - k'), \quad (9)$$

with $f_{ab}(k) = \int_0^L dx e^{-ikx} w_a(x) w_b(x)$.

The first-order momentum-space correlation is

$$G_1^{ab}(k, k') = \langle \hat{\psi}_b^\dagger(k') \hat{\psi}_a(k) \rangle = \langle \psi_b^*(k') \psi_a(k) \rangle_W - \frac{1}{2} f_{ab}(k - k'), \quad (10)$$

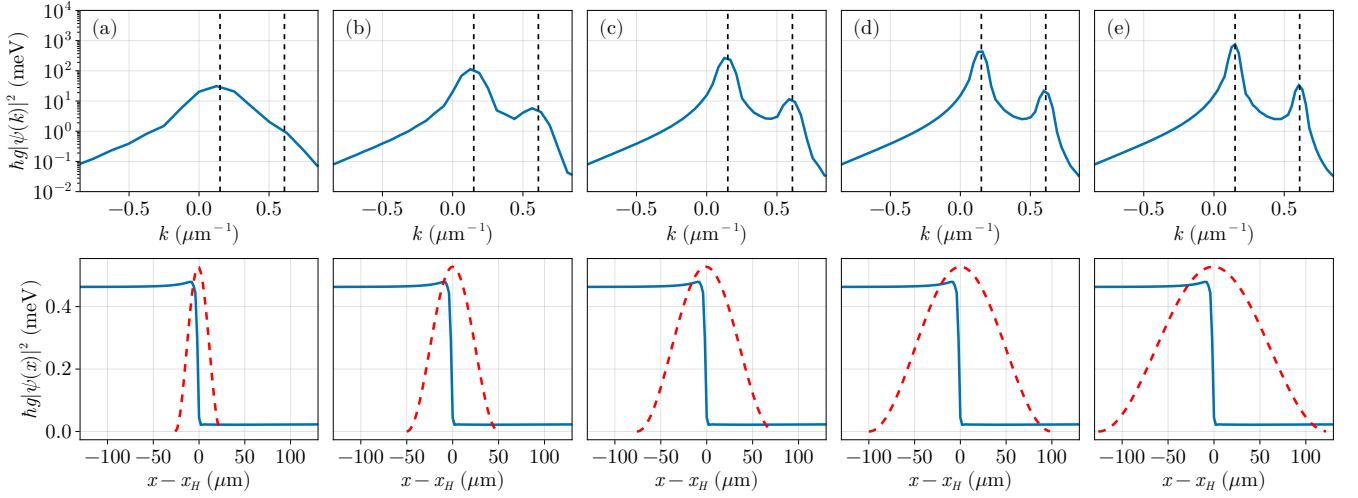


FIG. 2. Bottom row: different window widths shown over the mean-field position density. Top row: corresponding mean-field momentum density. Dashed red lines: windows. Black vertical lines: pump wavenumbers k_{up} and k_{down} .

with populations $n_a(k) = G_1^{aa}(k, k)$. The cross-region second-order correlation is

$$\begin{aligned}
 G_2(k, k') &= \langle \hat{\psi}_u^\dagger(k') \hat{\psi}_d^\dagger(k) \hat{\psi}_d(k) \hat{\psi}_u(k') \rangle \\
 &= \langle |\psi_u(k')|^2 |\psi_d(k)|^2 \rangle_W \\
 &\quad - \frac{1}{2} \langle |\psi_u(k')|^2 \rangle_W f_{dd}(0) \\
 &\quad - \frac{1}{2} \langle |\psi_d(k)|^2 \rangle_W f_{uu}(0) \\
 &\quad - \text{Re} [\langle \psi_u^*(k') \psi_d(k) \rangle_W f_{ud}(k - k')] \\
 &\quad + \frac{f_{uu}(0) f_{dd}(0)}{4} \\
 &\quad + \frac{|f_{ud}(k - k')|^2}{4}.
 \end{aligned} \tag{11}$$

The normalized correlation is $g_2(k, k') = G_2(k, k') / [n_u(k') n_d(k)]$.

SYMMETRY PROPERTIES OF $\delta g_2(k, k')$

Two symmetries constrain the correlation maps.

Exchange symmetry with equal overlapping windows. When both windows are identical and centered on the horizon [Fig. 2(a) of the main text], $\delta g_2(k, k') = \delta g_2(k', k)$. This should not be interpreted as a $q \leftrightarrow q'$ symmetry: in our convention $k = k_p + q$, a Bogoliubov mode at $+q$ contains field components at $\pm q$ with unequal weights $u_q \neq v_q$, so generically $\delta g_2(k_p + q, k') \neq \delta g_2(k_p - q, k')$. With separated windows this exchange symmetry is broken.

Sign-reversal symmetry. The global transformation $(q, q') \mapsto (\pm q, \pm q')$ is exact, enforced by the real nature of the field operator and the bosonic commutation relations. Each physical trace (HR-dn* or HR-down) therefore appears in four quadrants, yielding eight traces total. This serves as a consistency check [9, 12].

WINDOW PLACEMENT CONSIDERATIONS

The use of a windowed Fourier transform serves several purposes. Physically, placing two windows on opposite sides of the horizon isolates the cross-horizon Hawking signal from autocorrelations. Technically, the numerical algorithm assumes periodic boundary conditions, requiring modifications of the pump and losses near the edges (see Sec.); the windows exclude these artifacts from the momentum-space calculation.

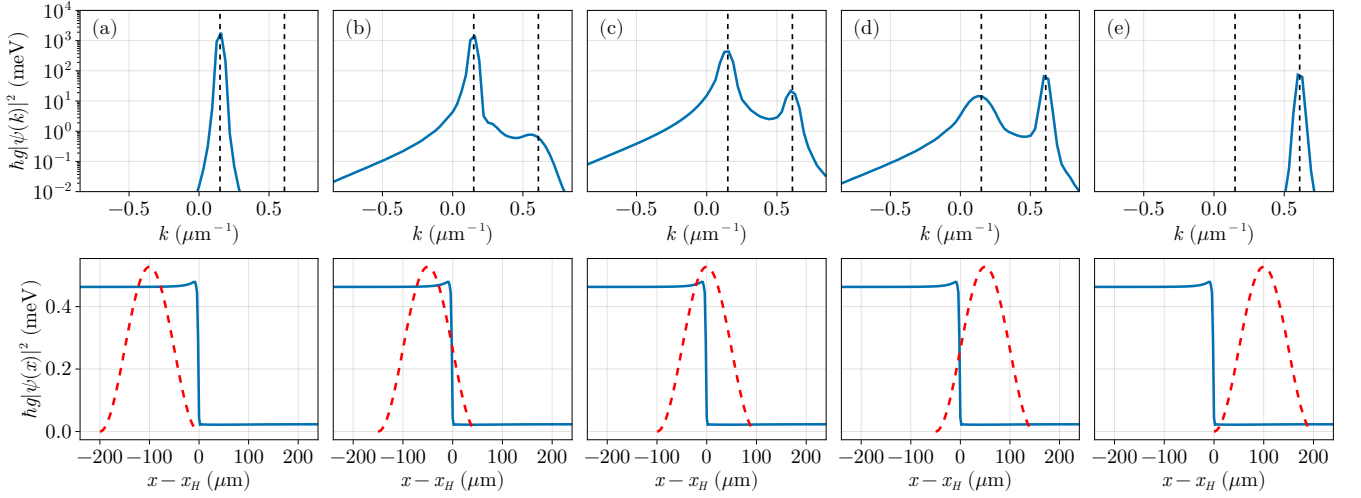


FIG. 3. Bottom row: different window placements shown over the mean-field position density. Top row: corresponding mean-field momentum density. Dashed red lines: windows. Black vertical lines: pump wavenumbers k_{up} and k_{down} .

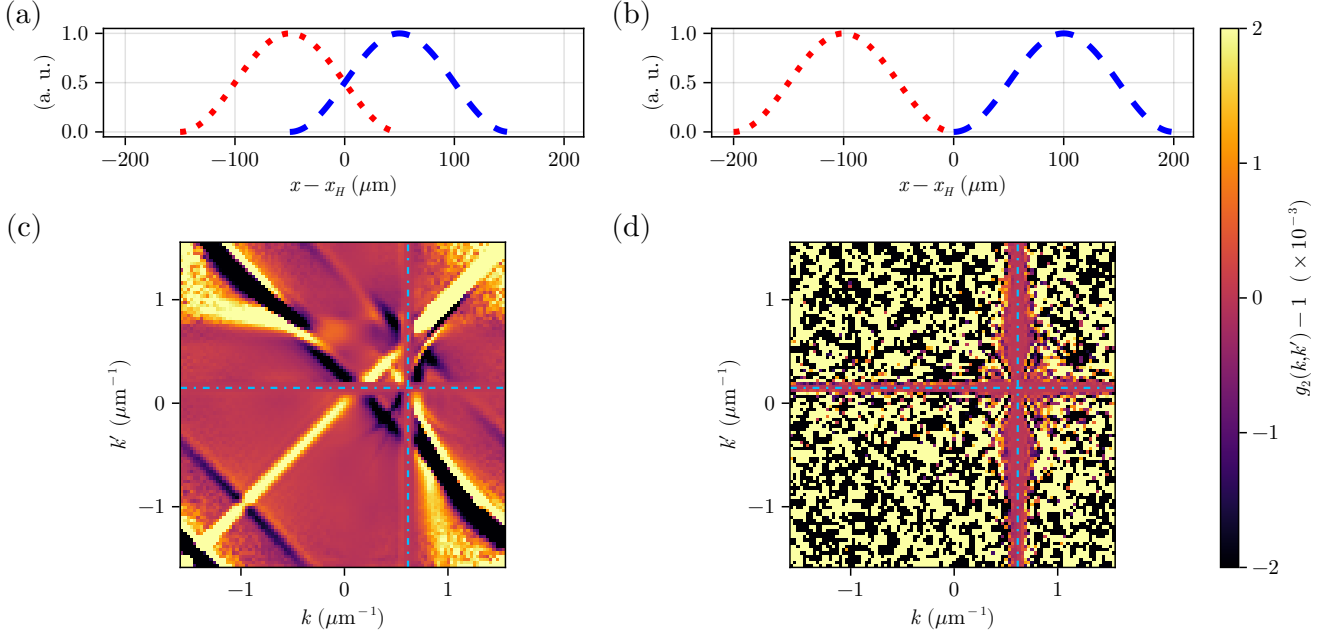


FIG. 4. (a) and (b): pairs of windows that generate the momentum density correlations in (c) and (d), respectively. In this simulation, 1.7×10^8 realizations were performed.

Proper use of window functions requires consideration of both size and placement:

Window size and spectral resolution. The window length L_w defines the momentum resolution $\Delta k_w = 2\pi/L_w$ —a manifestation of the uncertainty relation between Fourier conjugates. A localized window gives poor momentum resolution. Fig. 2 illustrates this: the momentum density $|\psi(k)|^2$ for windows of decreasing length shows the upstream and downstream pump peaks merging due to spectral leakage.

Window placement and polariton lifetime. One would ideally place windows fully on opposite sides of the horizon to eliminate autocorrelations [Fig. 3(a,e)]. However, this leaves only the sharp pump peaks in the momentum density because the Hawking excitations—created at the horizon—decay before reaching the window maximum. This is a direct consequence of the finite polariton lifetime and contrasts with the situation in closed (conservative) systems [9].

The optimal compromise, employed in the main text, is to place the windows partially overlapping the horizon [Fig. 3(b,d)].

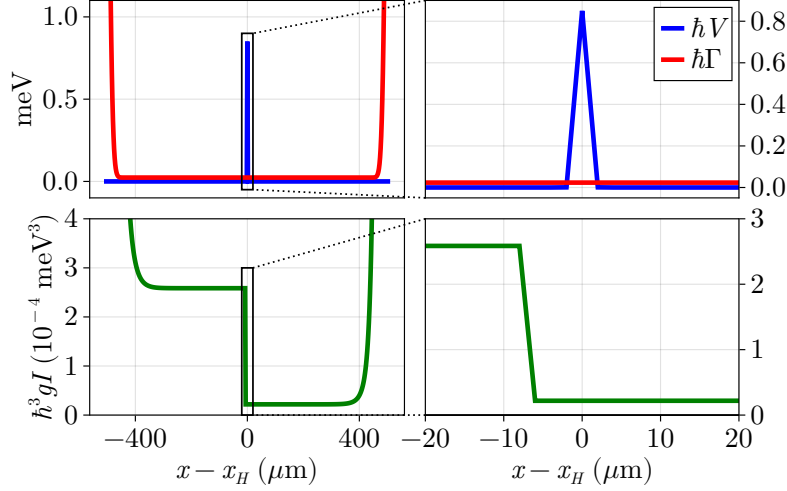


FIG. 5. Top row: potential and losses; bottom row: intensity profile.

This ensures sufficient momentum resolution, significant damping of autocorrelations, and allows the Hawking radiation to reach the window maximum before leaving the cavity.

Fig. 4 further illustrates this: when windows are fully contained within each region (b), the correlations (d) display only noise, consistent with negligible occupation of the relevant modes.

STABILIZATION OF THE FLUID IN THE NUMERICAL SIMULATION

Working close to the critical point of the bistability cycle makes stabilization challenging. The periodic boundary conditions of the numerical algorithm further complicate this. We use the following prescription.

Modified losses. A spatially dependent loss

$$\Gamma(x) = \gamma + \gamma_{damp}[G(x; 0, w_{damp}) + G(x; L, w_{damp})] \quad (12)$$

rises significantly near the edges, where $G(x; x_0, w) = e^{-(x-x_0)^2/w^2}$.

Modified pump. The pump is modified to rise at the edges:

$$F(x, t) = \begin{cases} [F_{up} + (F_M - F_{up}) \text{sech}(x/w_p)] e^{ik_{up}x - \omega_p t} h(t), & x < x_s \\ [F_{down} + (F_M - F_{down}) \text{sech}((x-L)/w_p)] e^{ik_{down}x - \omega_p t} h(t), & x > x_s \end{cases} \quad (13)$$

where x_s is the position of the pump step and $h(t) = 1 + A e^{-t/\tau}$ provides an initial amplitude boost to reach the top of the bistability curve, decaying after a transient time τ .

Localized potential. A Gaussian potential $V(x) = V_0 G(x; x_d, w_V)$ is centered at position x_d .

All terms are illustrated in Fig. 5.

NUMERICAL PARAMETERS

Table I lists all parameters used in the simulations.

FROM MOMENTUM-SPACE $g^{(2)}$ TO PHOTOCURRENT CORRELATIONS

Let the total optical power collected from window $a \in \{u, d\}$ be P_a . The mean photo-electron rate on detector a is

$$R_a = \eta_a \mathcal{T}_a \frac{P_a}{\hbar \omega_0}, \quad (14)$$

Parameter name	Symbol	Value
Cavity length	L	$1024 \mu\text{m}$
Grid points	N	512
Grid spacing	Δx	$2 \mu\text{m}$
Time step	Δt	0.2 ps
Duration	T	400 ps
Realizations	N_{sim}	5.5×10^8
Planck's constant	$\hbar$	$0.6582 \text{ meV} \cdot \text{ps}$
Loss rate	γ	$0.047 \text{ meV}/\hbar$
Polariton mass	m^*	$1/6 \text{ meV} \cdot \text{ps}^2 \cdot \mu\text{m}^{-2\text{a}}$
Detuning	$\delta(0)$	$0.49 \text{ meV}/\hbar$
Interaction constant	g	$3 \times 10^{-4} \text{ meV} \cdot \mu\text{m}/\hbar$
Edge damping	γ_{damp}	$4.5 \text{ meV}/\hbar$
Edge damping width	w_{damp}	$20 \mu\text{m}$
Upstream pump k	k_{up}	$0.148 \mu\text{m}^{-1}$
Downstream pump k	k_{down}	$0.614 \mu\text{m}^{-1}$
Upstream pump amplitude	F_{up}	$1.410 \text{ ps}^{-1} \cdot \mu\text{m}^{-1/2}$
Downstream pump amplitude	F_{down}	$0.410 \text{ ps}^{-1} \cdot \mu\text{m}^{-1/2}$
Max pump amplitude	F_{M}	$20 \text{ ps}^{-1} \cdot \mu\text{m}^{-1/2}$
Pump edge width	w_{p}	$20 \mu\text{m}$
Step position	x_{s}	$505 \mu\text{m}$
Initial boost factor	A	6
Transient time	τ	50 ps
Defect position	x_{d}	$512 \mu\text{m}$
Potential amplitude	V_0	$0.85 \text{ meV}/\hbar$
Potential width	w_{V}	$0.75 \mu\text{m}$

^aRoughly 3×10^{-5} the electron mass.

TABLE I. Numerical values of all simulation parameters.

with quantum efficiency η_a and optical throughput $\mathcal{T}_a$. The normally ordered cross-correlation of the photocurrents is

$$C_{ud}(\tau) \equiv \langle : \delta i_u(t) \delta i_d(t + \tau) : \rangle = e^2 R_u R_d [g_{ud}^{(2)}(\tau) - 1], \quad (15)$$

where $g_{ud}^{(2)}(\tau)$ is the time-domain second-order correlation integrated over the momentum windows.

In continuous-wave operation with sampling bin $\Delta t \sim \tau_c$ (set by the polariton lifetime and detection bandwidth), the zero-delay correlation is

$$g_{ud}^{(2)}(0) \simeq 1 + \frac{R_{\text{pair}}}{R_u R_d \Delta t}, \quad (16)$$

so $\delta g^{(2)} \equiv g_{ud}^{(2)}(0) - 1$ gives the pair rate

$$R_{\text{pair}} = \delta g^{(2)} R_u R_d \tau_c. \quad (17)$$

This allows interpreting the numerically obtained $\delta g^{(2)}$ as an excess coincidence probability per correlation time.

When $\delta g^{(2)} \ll 1$ (our case), accidental coincidences dominate the variance:

$$\text{SNR} \simeq \delta g^{(2)} \sqrt{R_u R_d \tau_c T}. \quad (18)$$

A significance of s standard deviations requires

$$T \simeq \frac{s^2}{[\delta g^{(2)}]^2 R_u R_d \tau_c}. \quad (19)$$

For illustration: with $R_u = R_d = 10^8 \text{ s}^{-1}$ (from $P_a \sim 0.1 \mu\text{W}$ at near-IR frequencies), $\tau_c \sim 10 \text{ ps}$, and $\delta g^{(2)} \sim 10^{-3}$, a 5σ detection requires $T \approx 250 \text{ s}$ (about four minutes).

Balanced detection suppresses classical common-mode noise from the pump: each momentum window is split 50:50, the photocurrents of each pair are subtracted, and the Hawking signal is obtained from the cross-correlation $\langle \tilde{i}_u(t) \tilde{i}_d(t + \tau) \rangle$. The

mean singles rates in Eq. (19) are then the post-split rates; the improvement comes from removing classical intensity noise without affecting the normally ordered cross-correlation.

Alternatively, one may estimate the zero-frequency cross-spectrum $S_{ud}(0)$ using RF spectrum analyzers with bandwidth B . The integrated SNR scales identically to Eq. (19). Fast time resolution is unnecessary provided the electronics bandwidth exceeds $1/\tau_c$.

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